



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
REVIEW TEST
04-09-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TNAS]

Time : 3 Hours

PAPER-2

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

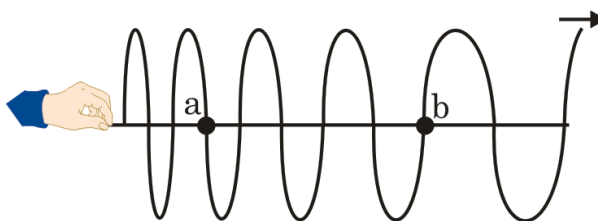
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

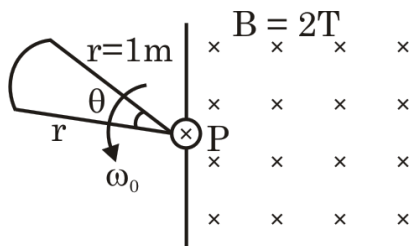
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. A boy is jerking one end of a uniform taut string. The wave train propagating to the right has been shown in the figure.

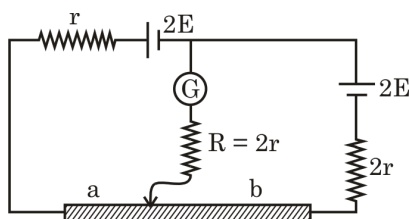


- (A) The boy is jerking at gradually decreasing frequency.
- (B) The boy is jerking at gradually increasing frequency.
- (C) Speed of particle at a is more than speed of particle at b.
- (D) Speed of particle at b is more than speed of particle at a.

2. A conducting loop has two straight and a circular arc sections. It is hinged at point P and given $\omega_0 = 2 \text{ rad/s}$ angular velocity. Angle of circular arc is $\frac{\pi}{4}$ rad and radius 1m. Loop has total resistance 1Ω and moment of inertia $\frac{\pi}{4} \text{ kgm}^2$. As loop enters in the magnetic field, induced current appears and hinge is insulator. Then :-

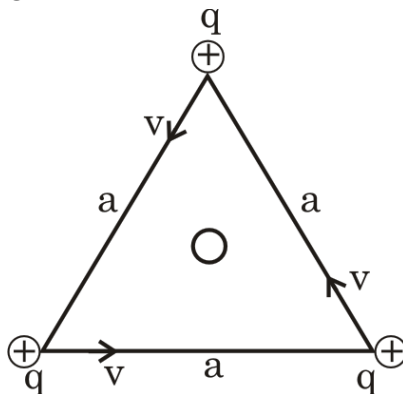


- (A) Loop rotates uniformly for time $\frac{\pi}{2}$ s in magnetic field.
- (B) Loop rotates uniformly for time $\frac{\pi}{4}$ s in magnetic field.
- (C) Loop is able to complete more than 2 cycles if initial angular velocity is 8 rad/s.
- (D) Loop is able to complete more than 2 cycles if initial angular velocity is 7 rad/s.
3. A meterbridge circuit is shown in figure. The resistive wire has a total resistance $6r$. Jockey divides the wire in two parts of length a and b .



- (A) For zero deflection in galvanometer $a : b = 7 : 5$
- (B) For zero deflection in Galvanometer $a : b = 4 : 3$.
- (C) If galvanometer is replaced by capacitor then for maximum time constant $a : b = 7 : 5$
- (D) If galvanometer is replaced by capacitor then for maximum time constant $a : b = 4 : 3$
4. A series R-L circuit and a series R-C circuit have same power factors $\frac{1}{\sqrt{2}}$. When these circuits are connected with same voltage source at different times, the maximum current of R-L circuit is found 9 times less of maximum current of R-C circuit. Now these circuits are connected in series together with same voltage source. Then :-
- (A) Final circuit is capacitive in nature. (B) Final circuit is inductive in nature.
- (C) Power factor of final circuit is 0.78. (D) Power factor of final circuit is 0.62.

5. Three charge particles are placed at vertices of equilateral triangle. They start moving such that they always follow each other in anticlockwise manner along sides of triangle with uniform speed v . There is a superconducting ring at center of triangle having radius r . The nature of ring is such that its flux always remains constant with help of induced current. Ring is very small in size and charges were initially very far. Initially there is no current in ring.



- (A) When side of triangle is a the current in ring is $\frac{9qvr}{4\pi a^2}$.
- (B) When side of triangle is a the current in ring is $\frac{5}{4} \frac{qvr}{\pi a^2}$.
- (C) Magnetic dipole moment of ring varies linearly with time.
- (D) Magnetic force on any charge due to ring varies inversely with 5^{th} power of triangle's side.
6. A transparent sphere of refractive index 1.5 has radius R . There is a point light source S inside the sphere at some distance from center. A person observes the light source along diametric line passing through S . Source appears at $\frac{4}{7}R$ distance from surface.
- (A) Source is at $\frac{2R}{3}$ distance from center of sphere.
- (B) When person observes the source from diametrically opposite point it appears $\frac{8R}{5}$ distance from surface.
- (C) All light from source passes the spherical boundary without any total internal reflection.
- (D) Source is at $\frac{R}{3}$ distance from center of sphere

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen.

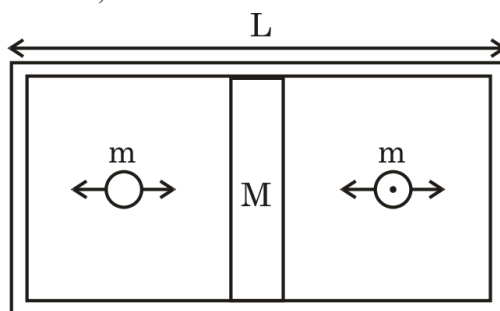
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

Consider a fixed highly evacuated cylinder in a gravity free region. Inside the cylinder piston of mass M can move without friction. In each part of the cylinder, a ball of mass m ($m \ll M$) is moving back and forth with very high speed v and in doing so they are colliding elastically with the ends of the cylinder and with the piston. When the piston is at the center, it is in equilibrium as shown in figure.

Take $M = 0.2 \text{ kg}$, $m = 10^{-3} \text{ kg}$, $L = 2 \text{ m}$, $v = \pi \text{ m/s}$



7. Now the piston is shifted slightly towards one of the ends of cylinder with a negligible speed and released. Time period of oscillations of piston is :-
- (A) 1.5 (B) 20
(C) 3.2 (D) 45
8. If the ball on right side has double mass as well as speed of left side ball then length of right section of cylinder in steady state is :-
- (A) $\frac{4L}{5}$ (B) $\frac{8L}{9}$
(C) $\frac{7L}{8}$ (D) $\frac{3L}{4}$

Paragraph for Questions 9 and 10

In resonance column experiment a tuning fork of frequency $f = 400$ Hz is held above the pipe as shown in figure. The reservoir is raised and lowered to change the level of water and thus the length of the column of air in the tube. The area of cross section of the reservoir is 6 times that of the pipe. Initially, the reservoir is kept so that the pipe is full up to the brim. Tuning fork is sounded and the reservoir is lowered. When the reservoir is lowered by 21 cm, first resonance is recorded. When the reservoir is lowered further by 49 cm the second resonance is heard.



9. The area of reservoir is :-

(A) $54 \pi \text{ cm}^2$

(B) $112 \pi \text{ cm}^2$

(C) $27 \pi \text{ cm}^2$

(D) $150 \pi \text{ cm}^2$

10. The speed of sound is :-

(A) 228 m/s

(B) 336 m/s

(C) 398 m/s

(D) 328 m/s

SECTION-II (i) : (Maximum Marks: 12)

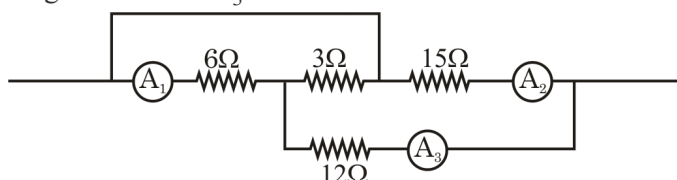
- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

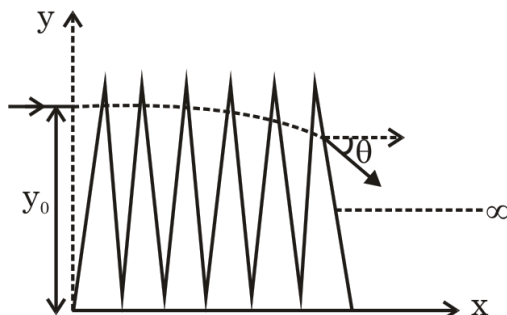
A portion of a large network is shown in figure. Values of resistances are indicated and all ammeters are ideal. In steady state reading of ammeter A_3 is 3A.



1. Reading of ammeter A_1 is A.
2. Reading of ammeter A_2 is A.

Paragraph for Questions 3 and 4

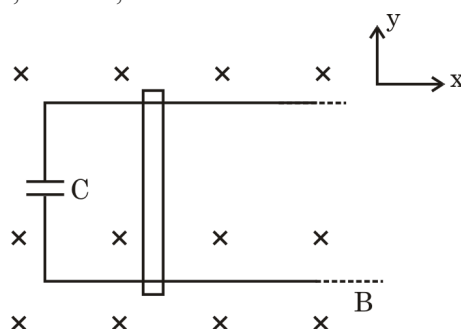
Large number of identical thin prisms are placed on x-axis. Each has refractive index 1.5 and angle of prism $\frac{\pi}{2} \times 10^{-3}$ rad. A light ray incidents parallel to x-axis along line $y = y_0$. Take thickness of each prism $t = 1$ mm. Take $\ln 2 = 0.7$.



3. When ray reaches $x = 1$ m angle made by ray with x-axis is θ rad. Value of θ is
4. When ray reaches $x = 1$ m its y displacement magnitude in meter is

Paragraph for Questions 5 and 6

An infinite conducting wire is fixed in gravity free space with a capacitor C. A conducting rod of mass M and length ℓ can also slide on wire frame without any friction system is placed in uniform magnetic field B . Resistance of wire as well as rod is negligible and rod is initially at rest with respect to wire frame. Now rod is imparted an impulse towards positive x axis $\vec{J} = J\hat{i}$ at $t = 0$. Take $J = 4$ kg m/s, $M = 1$ kg, $\ell = 0.5$ m, $B = 2$ T, $C = 2$ F.



5. Displacement of rod in $t = 1$ s is
6. Energy lost in circuit till steady state is achieved is

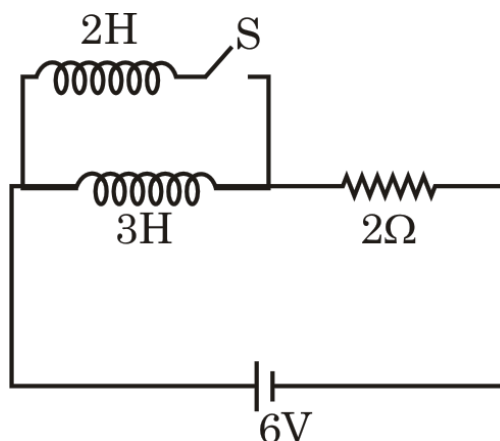
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
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- For each question, marks will be awarded in one of the following categories :

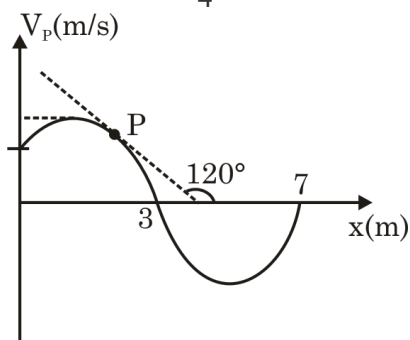
Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. In the circuit shown in figure. Switch is initially open for long time and at $t = 0$ it is closed. Current in switch at $t = 16$ s is $\frac{N}{3}$. Find value of N. Ignore the mutual inductance.



8. A travelling wave is propagating along a tight rope. Velocity of any point of rope is shown in the graph. There is a point P on rope. If frequency of wave is $\frac{\sqrt{3}}{4}$ Hz then magnitude of acceleration of P is ?



9. A string of mass 1gm, length 3m, tension 30 N is fixed at both the ends and is oscillating in 3rd overtone. The amplitude at the antinode is 0.1 mm. What is the energy of oscillations (in μ J) ?

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
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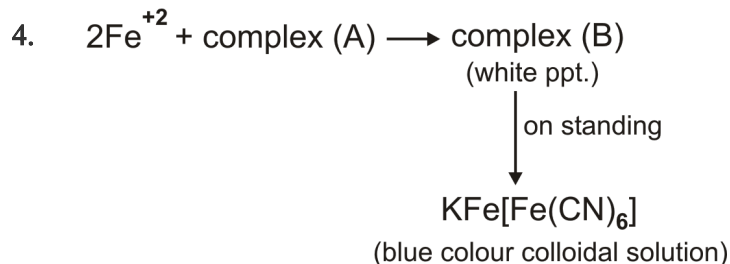
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- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Compressibility of real gas will be less than ideal gas :
 - (A) At low pressure when, temperature of gas < Boyle's temperature
 - (B) At low pressure when, temperature of gas > Boyle's temperature
 - (C) At very high pressure when, temperature of gas < Boyle's temperature
 - (D) At very high pressure when, temperature of gas > Boyle's temperature
2. The initial rate of hydrolysis of 0.2M methyl acetate by a strong acid HA (0.1 M) at 25°C is 0.1 M sec^{-1} . While the initial rate of hydrolysis of 0.4 M methyl acetate by a weak acid HB (1M) at 25°C is $10^{-3} \text{ M sec}^{-1}$. Select the correct option(s). (Given : $\ln 2 = 0.7$)
 - (A) $K_{a \text{ HB}} = 2.5 \times 10^{-7}$
 - (B) First order rate constant of hydrolysis reaction will be higher when acid HB is used.
 - (C) First order rate constant of hydrolysis reaction will be higher when acid HA is used.
 - (D) Half life when acid HB is used is 280 seconds.

3. Yellow solution of compound (A) is filtered and acidified with sulphuric acid to give a solution from which orange colour compound (B) can be crystallised. Choose the correct option(s).

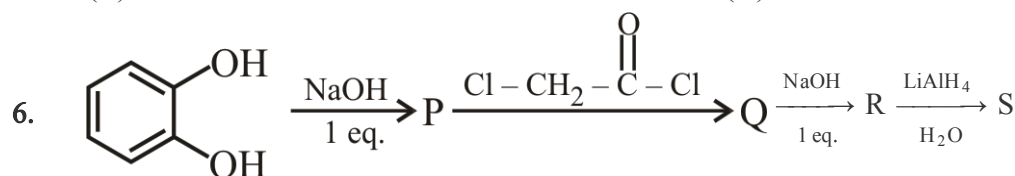
- (A) Compound A is tetrahedral
 (B) Compound B contains two tetrahedral units sharing one corner
 (C) Oxidation state of metal in compound (A) & (B) is same
 (D) All M–O bond length in compound (B) are equal to 179 pm



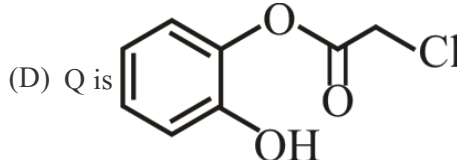
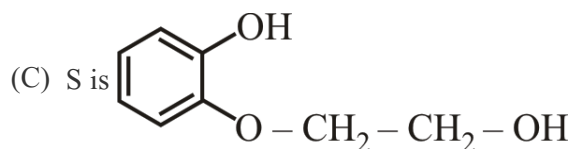
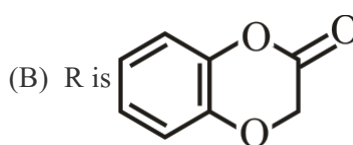
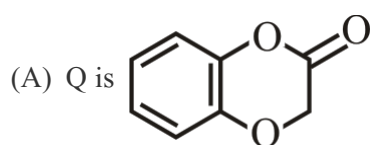
Choose the correct statement(s).

- (A) Hybridisation of central metal atom in complex (A) is d^2sp^3
 (B) Oxidation state of central metal atom in complex (B) is II
 (C) Complex (A) is colourless and diamagnetic in nature
 (D) Complex (A) has CFSE (Δ_0) value greater than Pairing energy
5. Which of the following acid remains unaffected on heating?

- (A) Fumaric acid
 (B) Malonic acid
 (C) Maleic acid
 (D) Succinic acid



Select correct statement :



SECTION-I (ii) : (Maximum Marks: 12)

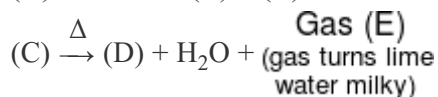
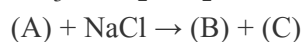
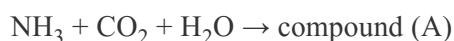
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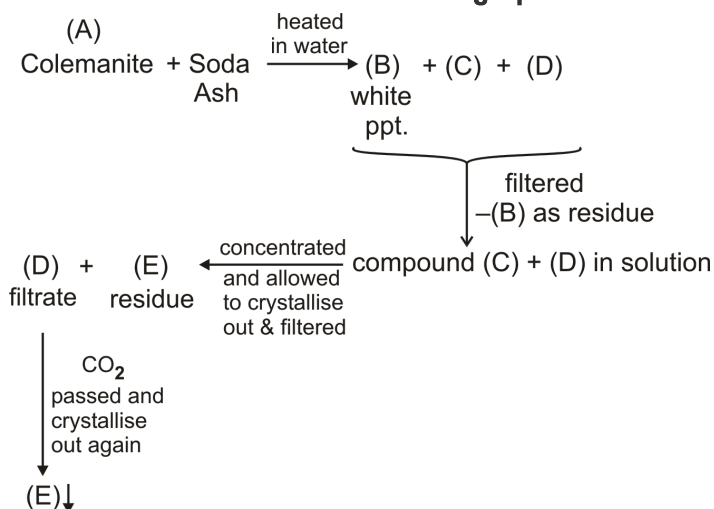
Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8



7. Choose the correct option for compound (D).
- (i) Produces green residue on fusion with CrCl_3 and NaNO_3
 - (ii) Produces white ppt. with BaCl_2 solution
 - (iii) Produces red brown ppt. with HgCl_2 solution
- (A) (i), (ii), (iii) are correct (B) only (i) and (ii) are correct
(C) only (i) and (iii) are correct (D) only (ii) and (iii) are correct
8. Compound (B) is used as auxiliary reagent for the group_____of cation. Which of the following cation belongs to this group?
- (A) Pb^{2+} (B) Al^{+3} (C) Ag^+ (D) Bi^{+3}

Paragraph for Questions 9 and 10



9. What is number of water of crystallisation in compound (E)?
- (A) 24 (B) 8
(C) 7 (D) 0
10. Acid produced by dissolving compound (C) in dil. HCl, when treated with H_2O_2 and NaOH forms compound (G). Choose the correct option regarding compound (G).
- (i) One of the atom has +3 oxidation state
(ii) It contains 2 peroxy linkage
(iii) All oxygen atoms have same oxidation number
- (A) only option (i) is correct
(B) only option (i) and (ii) are correct
(C) option (ii) and (iii) are correct
(D) All are correct

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

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Paragraph for Questions 1 and 2

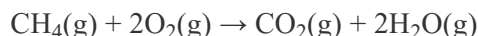
567.5 ml of HCl gas (measured at STP) is added to a 500 ml aqueous solution of strong base BOH containing 0.3075 g BOH.

The pH of resulting solution is found to be 1.4. If salt BCl forms NaCl type crystal having $r_{B^+} = 0.8\text{\AA}$ and $r_{Cl^-} = 1.2\text{\AA}$ ($\log 2 = 0.3$, $N_A = 6 \times 10^{23}$, Cl = 35.5)

1. Find relative atomic mass of atom B.
2. Find density of BCl crystal in g/ml.

Paragraph for Questions 3 and 4

Consider the combustion of 1 mole of CH_4 according to the following reaction with 20% excess air at 27°C and 1 bar pressure.



Assume air as a mixture of 20% O_2 and 80% N_2 by volume.

Given : $\Delta H^\circ_f, \text{CO}_2(\text{g}) = -100 \text{ K cal mol}^{-1}$

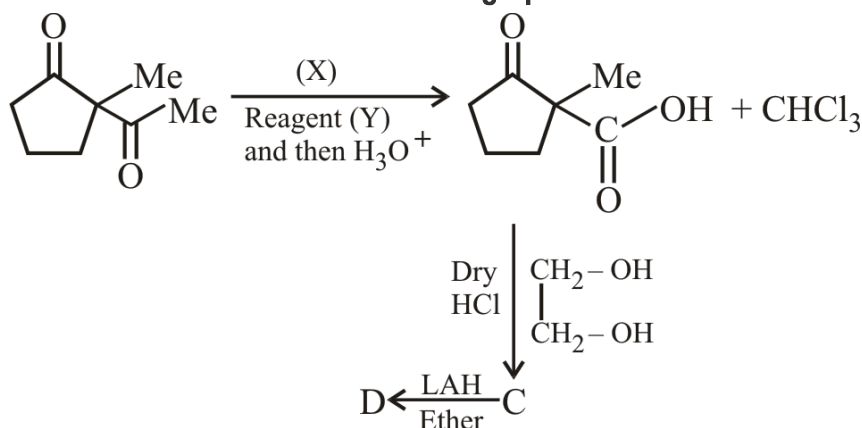
$\Delta H^\circ_f, \text{H}_2\text{O}(\text{g}) = -60 \text{ K cal mol}^{-1}$

$\Delta H^\circ_f, \text{CH}_4(\text{g}) = -20 \text{ K cal mol}^{-1}$

Assume all the gases are ideal gases and vibrational degree of freedom is not active. $R = 2 \text{ cal K}^{-1} \text{ mol}^{-1}$.

- Calculate maximum temperature in Kelvin that can be reached by above combustion. Ignore the variation of $\Delta_f H$ with temperature.
- If energy released by combustion of 1 mole of CH_4 at 127°C and 1 bar is supplied to a heat engine at 127°C which discards heat at 27°C and whose efficiency is 80% of maximum possible efficiency then calculate amount of heat discarded by the heat engine in Kcal. ($R = 2 \text{ cal K}^{-1} \text{ mol}^{-1}$).

Paragraph for Questions 5 and 6



- X react with hot and concentrated Y to give chloride and Z. Find total resonating structure of anion of Z.
- How many carbon present in principle carbon chain of product (D) according to IUPAC rule?

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. How many of the following will have yellow ppt. as the product.
- (i) $\text{Pb}^{2+} (\text{aq.}) + 2\text{NaOH} \rightarrow$
 - (ii) $\text{PbCrO}_4 + \text{dil. HNO}_3 \rightarrow$
 - (iii) $\text{Ag}_2\text{CrO}_4 + \text{excess NH}_3 \rightarrow$
 - (iv) $\text{Hg}^{2+} (\text{aq.}) + \text{NaOH} \rightarrow$
 - (v) $\text{K}_2[\text{Cd}(\text{CN})_4] + \text{H}_2\text{S} \rightarrow$
 - (vi) $\text{Ni}^{2+} (\text{aq.}) + \text{CN}^- (\text{excess}) \rightarrow$
 - (vii) $\text{Ca}^{2+} (\text{aq.}) + (\text{NH}_4)_2\text{CrO}_4 \xrightarrow{\text{CH}_3\text{COOH}}$
 - (viii) $\text{Ba}^{+2} (\text{aq.}) + (\text{NH}_4)_2\text{CrO}_4 \xrightarrow{\text{CH}_3\text{COOH}}$
8. How many isomers "X" of C_8H_{10} when reacts with hot alkaline KMnO_4 give only aromatic dicarboxylic acid?
9. How many Isomers Y of C_4H_8 when reacts with hot alkaline KMnO_4 give carbondioxide?

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Consider a square matrix A. Identify the correct statement -

- (A) If A is nilpotent matrix of index 2 then $A(I + A)^n = A \forall n \in \mathbb{N}$
 (B) If A is nilpotent matrix of index 2 then $A(I - A)^n = A \forall n \in \mathbb{N}$
 (C) If A is idempotent matrix, then $(I + A)^n = I + (2^n - 1)A \forall n \in \mathbb{N}$
 (D) If A is idempotent matrix, then $(I + A)^n = I - (2^n - 1)A \forall n \in \mathbb{N}$

2. If $z = \log \tan \frac{t}{2}$, and $\frac{d^2y}{dz^2} + 4y = 0$, then find the correct option

- (A) $\frac{d^2z}{dt^2} + \operatorname{cosec} t \cot t = 0$
 (B) $\frac{d^2y}{dt^2} + \cot t \frac{dy}{dt} + 4y \operatorname{cosec}^2 t = 0$
 (C) $\frac{d^2y}{dt^2} + \cot t \frac{dy}{dt} + 2y \operatorname{cosec} t = 0$
 (D) $\frac{d^3y}{dt^3} + \cos t \frac{dy}{dt} + y \cot t = 0$

3. Let $f(x) = \int_x^{x+\frac{\pi}{3}} |\sin \theta| d\theta$; where $x \in [0, \pi]$, which of the following is/are true
- (A) maximum value of $f(x)$ is 1
- (B) minimum value of $f(x)$ is $\frac{3}{2}$
- (C) maximum value of $f(x)$ is $\frac{1}{2}$
- (D) minimum value of $f(x)$ is $2 - \sqrt{3}$
4. Let f is a differentiable function satisfying $3f(x + 2y) = f(2x) + f(y) + 2x + 10y + 1 \forall x, y \in \mathbb{R}$ such that $f'(0) = 2f(0)$, then - (where C is constant of integration)
- (A) $f(x)$ is an odd function
- (B) $\int \frac{dx}{f(x)} = \ln(\sqrt{|f(x)|}) + C$
- (C) $f(0.5) = f'(5.0)$
- (D) $\sin(f(x))$ is a periodic function with period π
5. Choose the correct statement(s) -
- (A) The largest integer in the range of function $(\sin^{-1}x)^2 - (\cos^{-1}x)^2$ is 2
- (B) The smallest integer in the range of function $(\sin^{-1}x)^2 - (\cos^{-1}x)^2$ is -7
- (C) The value of 'x' satisfying $\sin^{-1}x + 2\cos^{-1}(-x) = 2\pi$, is greater than $\frac{1}{2}$
- (D) The value of 'x' satisfying $\sin^{-1}x + 2\cos^{-1}(-x) = 2\pi$, is less than $\frac{1}{2}$
6. Let $f_1 : (-\infty, 0) \rightarrow \mathbb{R}$, $f_2 : (-\infty, 0) \rightarrow \mathbb{R}$ and $f_3 : (1, \infty) \rightarrow \mathbb{R}$ be functions defined by $f_1(x) = e^x - \sqrt{-x}$, $f_2(x) = \sqrt{-x} + |x|$ and $f_3(x) = e^{1-x} - \sqrt{x-1}$. Then choose the correct option(s)
- (A) $f_1(x)$ is increasing function but $f_2(x)$ and $f_3(x)$ are decreasing functions
- (B) $f_1(x)$ is decreasing function but $f_2(x)$ is increasing function
- (C) $f_2(x)$ and $f_3(x)$ are decreasing functions
- (D) $f_2(x)$ is decreasing function but $f_1(x)$ and $f_3(x)$ are increasing functions

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

Let $f(x)$ be a differentiable function such that $f'(x) = \sqrt{2f(x) - f^2(x)}$, where $f'(0) = f(0) = 1$

7. Area bounded by $y = f(x)$ and x -axis from $x = 0$ to $x = \pi$ is

(A) $2\pi + 4$ (B) $\pi + 2$ (C) $\pi + 4$ (D) $2\pi + 2$

8. $\int_0^{\pi/2} \frac{f(x) - 1}{\cos^2 x + 1} dx$ is

(A) $\frac{\pi}{4}$ (B) $-\frac{\pi}{4}$ (C) $\frac{\pi}{2}$ (D) $\frac{3\pi}{4}$

Paragraph for Questions 9 and 10

Consider $f(x) = x^{\ln x}$ and $g(x) = e^2 x$. Let α and β ($\alpha < \beta$) be two values of x satisfying equation $f(x) = g(x)$.

9. Product of $\alpha\beta$ is -

(A) $\frac{1}{e^2}$ (B) $\frac{1}{e}$ (C) e (D) e^2

10. If $\lim_{x \rightarrow \beta} \frac{f(x) - c\beta}{g(x) - \beta^2}$ exists and equal to 'r' then value of $(c - r)$ is -

(A) $4 - e^2$ (B) $e^2 - 4$ (C) $4 - e$ (D) $e - 4$

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ satisfy $A^n = A^{n-2} + A^2 - I \forall n \geq 3$. Further consider a matrix U of order 3×3

with its column as U_1, U_2, U_3 such that $A^{50}U_1 = \begin{bmatrix} 1 \\ 25 \\ 25 \end{bmatrix}$; $A^{50}U_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$; $A^{50}U_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

1. Trace of A^{50} is
2. Sum of all entries of $UA^{50}U^{-1}$ is

Paragraph for Questions 3 and 4

Let $f(x) = 4^x$ and $g(x) = \sin(\cos(\cos\pi x) + \cos(\sin\pi x))$, then

3. The value of $\lim_{x \rightarrow \infty} \frac{g(x)}{f(x)}$ is equal to
4. $\int_0^4 f(x) \cdot g(x) dx = \lambda \int_0^{1/2} f(x) g(x) dx$, then λ is -

Paragraph for Questions 5 and 6

Let $y = f(x)$ be a fourth degree algebraic polynomial function such that

(i) $\lim_{x \rightarrow 1} \frac{f(x)}{(x-1)^2} = 2$

(ii) It cuts y-axis a point where slope of normal is $\frac{1}{8}$

(iii) Tangent at $x = 2$ is parallel to $y = 8x + 1$.

Consider $y = g(x)$ be a quadratic curve touching and cutting $y = f(x)$ on x-axis and y-axis respectively.

5. If angle of intersection between $y = f(x)$ & $y = g(x)$ at point where they cut y-axis is $\tan^{-1} \frac{a}{b}$ (where a and b are co-prime) then value of $(a + b)$ is -
6. The abscissae of point on curve $y = g(x)$, where length of subtangent is 2018 can be -

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. If area bounded on the left by y-axis, below by x-axis, right by $x = \frac{\pi}{2}$ above left by $y = \cos x$ and above right by $y = \sin x$ is K then value of $100K^2$ is -
8. Let A and B be two sets such that $n(A) = 4$ and $n(B) = 3$. If $(x, 1)$, $(y, 2)$, $(z, 1)$ and $(2, 3)$ are in $A \times B$, where x, y, z are distinct elements. If a function $f: A \rightarrow B$ defined & number of onto function from set A to set B is λ , then value of $\frac{\lambda}{6}$ is equal to
9. Let matrix $P = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$ and Q is another matrix such that $Q^T P = P^T$ and $\lambda Q^T = 2P^T - \sqrt{2}I$, then value of determinant of matrix $\text{adj}(I + Q^T) \cdot \text{adj}(\lambda P)$ is
(where I is unit matrix of order 2, $\text{adj}(X)$ and X^T denote adjoint and transpose of matrix X respectively)

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

Space for Rough Work